

Quantum Computing and the Future of Encryption: A Hybrid Solution

Bryce Vitale

Abstract—This research paper will explore how advances in quantum computing affects the security of modern cryptographic systems and how to combat that issue. The research will begin by reviewing current symmetric and asymmetric encryption methods and how they could be broken by Shor’s and Grover’s quantum algorithms. I will also compare the differences between classical and quantum computing to better understand why quantum computers can solve certain problems faster. I then plan to study post-quantum cryptographic techniques such as lattice, code, and hash-based algorithms and how they function. The goal of this paper is to gain a deeper understanding of how classical encryption can evolve and advance in the fast-paced modern world along with practical applications of securing emerging technology in the quantum realm. My solution is a simple hybrid encryption approach using classical encryption and post-quantum encryption. I will measure my success by comparing the theoretical security strength between classical and post-quantum encryption algorithms using existing research.

Keywords— *Grover’s algorithm; hybrid encryption; lattice-based cryptography; post-quantum cryptography; quantum computing.*

I. INTRODUCTION

Quantum computing changes the risk model for encryption. Even if a cryptographically relevant quantum computer is not available today, the possibility of a future break of RSA and elliptic curve cryptography changes how confidentiality should be treated now. Data that remains valuable for years can be recorded and stored today and decrypted later when quantum capability catches up. This “harvest now, decrypt later” problem makes encryption a durability requirement, not only a real time control [4].

The research problem in this paper is practical; How can modern systems transition toward post-quantum security without assuming that every endpoint can immediately support new algorithms? This would occur without placing all of one’s trust in a single newly standardized practice. Surveys and background work explain the threat posed by Shor’s and Grover’s algorithms and outline candidate post-quantum cryptographic families [1],[2],[5]. The gap is the reality of deployment. Many systems have long upgrade cycles and mixed client populations. Another issue is that many systems also have strict interoperability constraints. This makes a clean rollover rarely possible.

My approach is a hybrid key establishment design that uses classical and post-quantum mechanisms in parallel. This then then derives one symmetric session key for authenticated encryption. I call this construction the Dual Track Hybrid Envelope (DTHE). The key idea is simply that the post-quantum component hedges against Shor’s algorithm while the classical component preserves mature security. The compatibility aspect is key for perseverance while PQC continues to receive analysis. The result is a deployable design pattern with a clear security claim that the session key remains confidential if at least one of the two key establishment mechanisms remains secure. This is of course under the assumption that standard properties are transcript bound like HKDF and AEAD.

II. THREAT MODEL

Two quantum algorithms drive the cryptographic impact. Shor’s algorithm enables polynomial time factoring and discrete logarithms. This algorithm breaks RSA and standard elliptic curve systems at a fundamental level [2], [4]. Grover’s algorithm provides a quadratic speedup for brute force search, reducing effective security margins for symmetric primitives and hash functions. The typical ideology is to use larger parameters such as AES-256 instead of AES-128 [1], [4].

This paper assumes an adversary has two capabilities. First, the attacker can passively observe and record traffic today. The second is that the attacker may later obtain a quantum computer able to break classical public key cryptography. The attacker may also be active during the handshake. There they would be attempting to downgrade negotiation or replay handshake. The security goals include long term confidentiality of session keys and protected application data. Another security goal is for resistance to downgrade attacks during algorithm migration.

A useful urgency model is in Mosca’s timeline. If the time to migrate (Y) plus the required confidentiality lifetime of data (X) exceeds the time until a quantum break (Z), then an organization is already late [4]. This is true even if Z is uncertain because X and Y are often large in practice for

regulated environments. This motivates hybrid or post-quantum deployment.

III. RELATED WORK

Post-quantum cryptography research proposes families believed to resist known quantum attacks. These including lattice-based, code-based, and hash-based designs [3],[5]. Lattice-based approaches have become leading candidates due to a strong balance of security arguments and which increases performance [3],[5].

Standardization work has moved PQC from theory to deployment. NIST has finalized standards for both key establishment and signatures. FIPS 203 specifies ML-KEM (based on CRYSTALS-Kyber) for key encapsulation. FIPS 204 specifies ML-DSA is used for digital signatures [6], [12]. NIST has also emphasized that these primitives are intended to be combined with symmetric encryption for practical data protection after a shared secret is established [10].

Hybrid deployment is an active focus in the protocol community. IETF drafts discuss hybrid key exchange for TLS 1.3 and provide guidance on combining classical ECDH. For example, X25519 combined with Kyber-derived KEMs. The primary design intent is that the combined exchange remains secure if either component exchange remains secure, assuming a correct combiner and safe negotiation [8], [9]. On the operational side, NSA's CNSA Suite 2.0 guidance reflects that migration planning is already underway and highlights approved algorithm directions for national security systems [7], [11].

IV. HYBRID APPROACH (DUAL TRACK HYBRID ENVELOPE)

A. Design Goals

DTHE is designed around four goals. The first is that it must provide confidentiality if either the classical or post-quantum key establishment remains secure. The second goal is that it must keep the application layer stable. This means the system still produces a single symmetric key that can be used with standard AEAD. Thirdly, it must prevent downgrade and misbinding by binding key derivation to the authenticated handshake transcript. The last goal is that it must be cryptographically fluid. This ensures the system can transition from hybrid to post-quantum only operation without redesigning message formats or storage encryption.

B. Structure

DTHE separates key establishment from bulk encryption. A client and server perform two parallel key establishment steps:

1) Classical AKE: A standard authenticated key exchange produces a classical shared secret s_c . (such as TLS 1.3 with X25519)

2) Post-quantum KEM: ML-KEM-768 produces a post-quantum shared secret s_p and a KEM ciphertext c_p .

The two secrets are combined using HKDF-Extract with a salt derived from the transcript hash T , then expanded to a single session key K .

K is used with AEAD to encrypt and authenticate application data.

C. Algorithm Pseudocode

Inputs: transcript, hash T , suite id

Classical: s_c ($AKE_SharedSecret()$)

PQC: (c_p, s_p) ($ML-KEM.Encaps(ek_p_peer)$)
salt ($H(T)$)

PRK ($HKDF-Extract(salt, s_c || s_p)$)

K ($HKDF-Expand(PRK, "DTHE session key")$)

Use K for AEAD encryption (AES-256-GCM)

V. VALIDATION

DTHE relies on a standard hybrid composition claim that is used in current protocol discussions. This is if the combiner is designed correctly that the resulting session key remains confidential if at least one of the two component exchanges remains secure [8], [9]. The goal is risk reduction during migration, not an unrealistic claim of absolute immunity.

Claim 1 (secure confidentiality): Let s_c be the classical shared secret and s_p be the ML-KEM shared secret. Assume that at least one of $\{s_c, s_p\}$ is indistinguishable from random to the attacker. Assume HKDF behaves as a pseudorandom function when keyed with an unknown secret input. Then the derived session key K is indistinguishable from random, and AEAD security reduces to standard symmetric assumptions.

Claim 2 (downgrade resistance): If the transcript includes the negotiated algorithms and key shares and the handshake authentication covers that transcript, then an active attacker cannot force a silent fallback to classical only mode without detection. In DTHE, classical only operation is treated as a distinct negotiated mode, not an implicit behavior.

Limitations are practical. Post-quantum implementations are newer and may be more exposed to side channel leakage or subtle bugs than mature elliptic curve implementations. DTHE assumes standard hardening. This includes constant time code, validated randomness, and strict input checks. FIPS 203 explicitly notes input validation requirements for ML-KEM usage [6].

VI. PERFORMANCE

Hybrid designs trade bandwidth for risk reduction. ML-KEM sizes are larger than classical elliptic curve key shares. For ML-KEM-768, FIPS 203 lists an encapsulation key of 1184 bytes, a ciphertext of 1088 bytes, and a 32-byte shared secret [6]. In contrast, X25519 uses a 32-byte key share. The hybrid overhead is mainly the added KEM public key and ciphertext in the handshake.

<i>Mechanism</i>	<i>Public Key</i>	<i>Ciphertext</i>	<i>Migration</i>
X25519 ECDH	32 B	32 B	Widely deployed. Broken by Shor's
ML-KEM-768	1184 B	1088 B	NIST FIPS 203 KEM
DTHE (hybrid)	32 B + 1184 B	32 B + 1088 B	Secure if either component holds.

^a. Representative Sizes

Fig. 1. Key Establishment Overhead

In a TLS like handshake where one side sends an ML-KEM encapsulation key and the peer returns an ML-KEM ciphertext, the post-quantum addition is about 2272 bytes plus framing. For most broadband and data center environments that overhead is manageable. It matters more on constrained links and high loss mobile networks. In those cases, session resumption and caching become more important.

Cryptographic versatility is not just a design preference. It is an operational requirement. A hybrid design should support switching parameter sets, removing classical components once the ecosystem is ready, and rotating keys without changing application encryption formats. This is also where policy pressure shows up. That government and regulated environments will require inventory and migration plans to validate implementations that match published guidance [7], [11].

VII. EXPERIMENT

A proof concept implementation is a realistic validation path for this approach. The prototype can implement three modes. Those being classical only (X25519), post-quantum only (ML-KEM), and hybrid (DTHE). For each mode, the system should measure total handshake bytes and handshake latency. Additionally, measuring CPU time per handshake and failure rate under packet loss is critical. The test should run under at least two environments. One of which is a normal LAN. The other is a simulated constrained network.

The expected result is that hybrid mode increases handshake size but keeps compute costs acceptable while materially reducing long term confidentiality risk. A secondary expected result is that transcript binding and strict negotiation are necessary to avoid downgrade paths during staged rollout.

VIII. CONCLUSION & FUTURE WORK

Quantum computing changes the long term meaning of encryption. Migration takes time, and data can remain valuable for years. This means the transition to post-quantum security is already relevant. This paper proposed DTHE, a dual-track hybrid envelope that combines classical authenticated key exchange with ML-KEM and derives a single transcript bound symmetric session key for AEAD. The design is intentionally practical and aligns with how modern systems already operate. Public key mechanisms establish secrets while symmetric encryption protects real data.

Future work is mainly engineering and deployment. The first step is post-quantum signatures and certificate chains will need to replace classical ones over time [12]. This is consistent with ML-DSA standardization. The next step is implementations should be validated and hardened against side channels. Lastly, organizations should build cryptographic versatile configuration and inventory practices to meet migration timelines, which are reflected in current guidance [7], [11].

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